

Lab 3: Verification of the Maximum Power Transfer Principle

ESC 221 Basic Principles of Electrical Engineering

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I. Introduction, Experimental Method and Hypotheses

A. Introduction

This experiment aims to verify the maximum power transfer principle, which states that maximum power is delivered to a load when the load resistance equals the Thevenin equivalent resistance of the source.

B. Experimental Method

A DC circuit was constructed with a fixed voltage source and a resistor network. A potentiometer was used as a variable load resistor to adjust the load resistance. The voltage and current were measured across the load resistor for varying resistance values from $500\ \Omega$ to $5000\ \Omega$ in increments of $250\ \Omega$. These measurements were used to calculate the power delivered to the load and identify the resistance value at which maximum power transfer occurs.

C. Hypotheses

Shown by calculations in the Appendix, the Thevenin equivalent resistance of the source is approximately $2.779\ \text{k}\Omega$. Therefore, it is hypothesized that maximum power transfer will occur when the load resistance is approximately equal $2.779\ \text{k}\Omega$ as shown in the theoretical plot of power vs. load resistance in Figure 1.

$R_3\ (\Omega)$	$V_3\ (\text{V})$	$I_3\ (\text{mA})$	$P_3\ (\text{mW})$
500	1.08	2.16	2.34
750	1.51	2.01	3.03
1000	1.88	1.88	3.53
1250	2.20	1.76	3.88
1500	2.49	1.66	4.12
1750	2.74	1.57	4.30
2000	2.97	1.48	4.41
2250	3.17	1.41	4.48
2500	3.36	1.34	4.52
2750	3.53	1.28	4.53
3000	3.68	1.23	4.52
3250	3.82	1.18	4.50
3500	3.96	1.13	4.47
3750	4.08	1.09	4.43
4000	4.19	1.05	4.38
4250	4.29	1.01	4.33
4500	4.39	0.975	4.28
4750	4.48	0.942	4.22
5000	4.56	0.912	4.16

Table 1: Calculated power values for varying load resistance values.

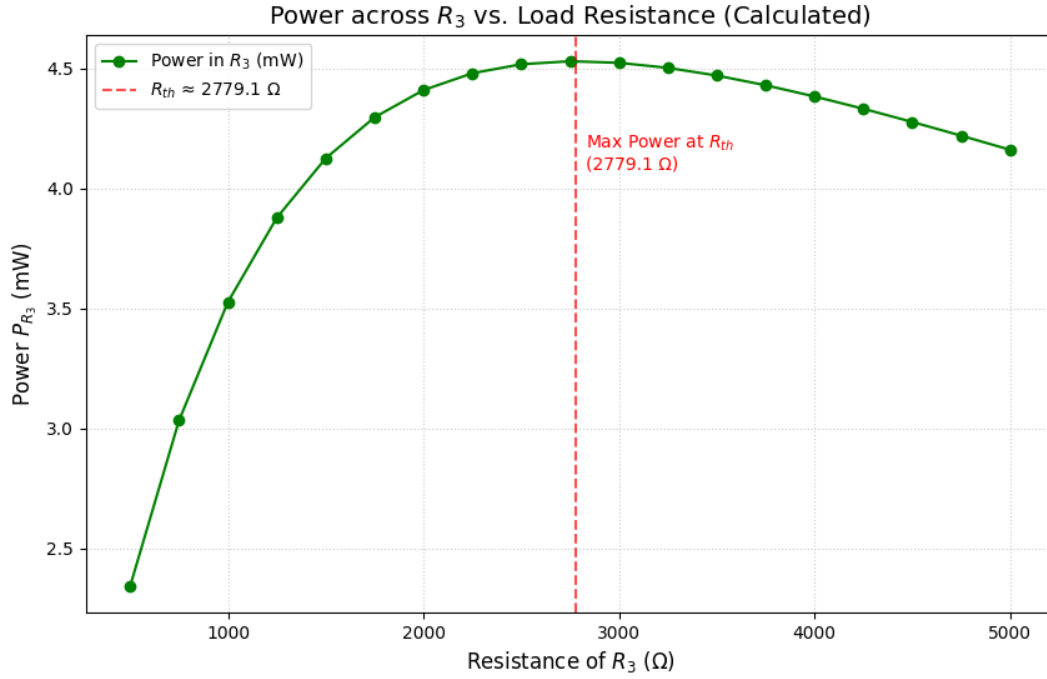


Figure 1: Theoretical power delivered to the load resistor vs. load resistance with max power transfer occurring at $R_3 = R_{th}$.

The plot of load voltage vs. load current (V_{R_3} vs. I_{R_3}) is expected to show the following linear relationship as shown in Figure 2:

$$V_{th} - I_{R_3}R_{th} - V_{R_3} = 0 \quad (\text{KVL})$$

$$V_{R_3} = V_{th} - I_{R_3}R_{th}$$

$$\Rightarrow \frac{\Delta V_{R_3}}{\Delta I_{R_3}} = -R_{th}$$

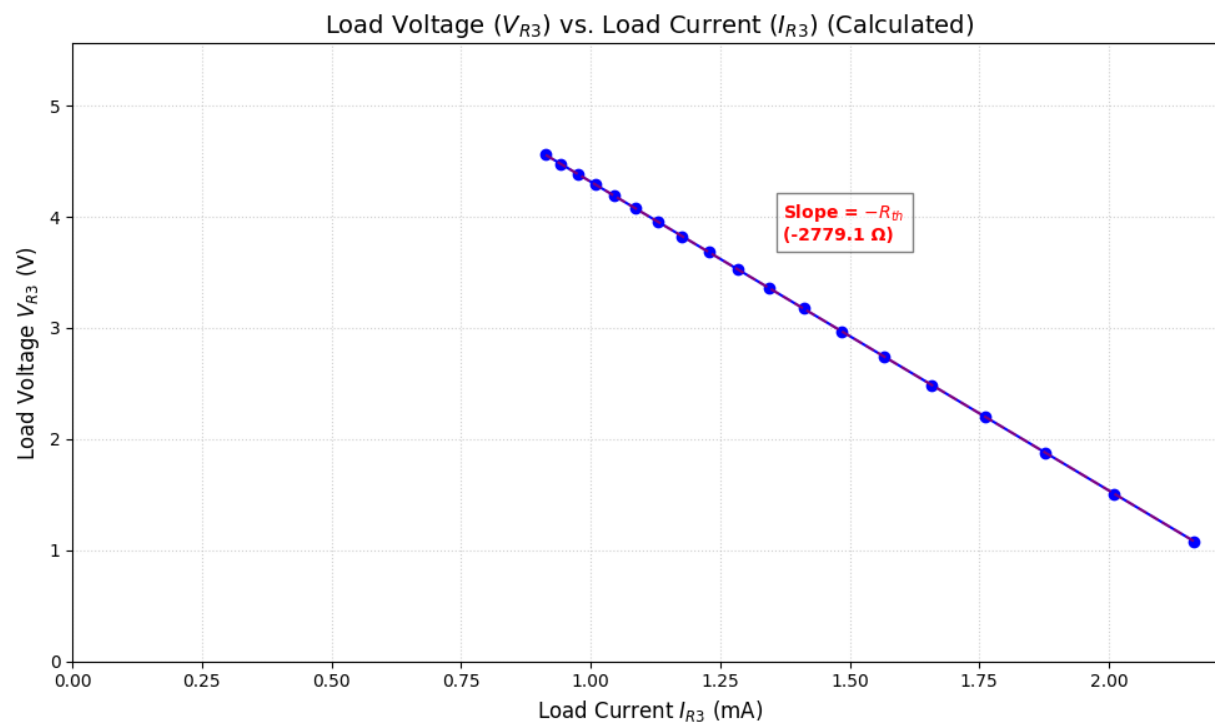


Figure 2: Theoretical load voltage vs. load current with slope of magnitude equal to the Thevenin equivalent resistance R_{th} .

II. Equipment List

1. **Digital Multimeter**

Manufacturer: AstroAI

Model: AM33B

2. **DC Power Supply**

Manufacturer: GW Instek

Model: SPD-3606

III. Wiring / Schematic Diagram / Procedure

A. Procedure

1. Constructing the Circuit

1. The circuit was constructed as shown in Figure 3 with a fixed voltage source v_1 and resistors R_1 and R_2 with a variable load resistor R_3 connected across nodes a and b.
2. A multimeter was connected in parallel across the load resistor R_3 to measure the voltage V_{R_3} and resistance R_3 .
3. A multimeter was connected in series with the load resistor R_3 to measure the current I_{R_3} .

2. Data Collection

1. The load resistance R_3 was set to $500\ \Omega$ and confirmed using the multimeter.
2. The voltage V_{R_3} across the load resistor and the current I_{R_3} through the load resistor were measured and recorded.
3. Steps 1 and 2 were repeated for varying load resistance values from $500\ \Omega$ to $5000\ \Omega$ in increments of $250\ \Omega$.

B. Schematic Diagram

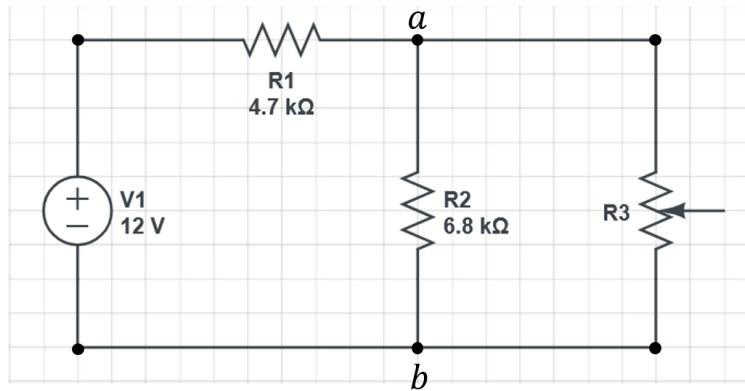


Figure 3: A DC circuit network consisting of a fixed voltage source v_1 and resistors R_1 and R_2 with a variable load resistor R_L connected across the output. Nodes a and b are labeled for measurement points.

IV. Results

R_3 (Ω)	V_3 (V)	I_3 (mA)	P_3 (mW)
500	1.34	2.68	3.60
750	1.85	2.46	4.54
1000	2.28	2.28	5.19
1250	2.65	2.12	5.62
1500	2.97	1.98	5.89
1750	3.25	1.86	6.07
2000	3.50	1.75	6.12
2250	3.72	1.65	6.15
2500	3.91	1.57	6.12
2750	4.09	1.49	6.07
3000	4.24	1.41	5.99
3250	4.38	1.35	5.89
3500	4.50	1.29	5.79
3750	4.61	1.23	5.67
4000	4.71	1.18	5.55
4250	4.80	1.13	5.42
4500	4.88	1.08	5.29
4750	4.95	1.04	5.16
5000	5.02	1.00	5.04

Table 2: Measured power values for varying load resistance values.

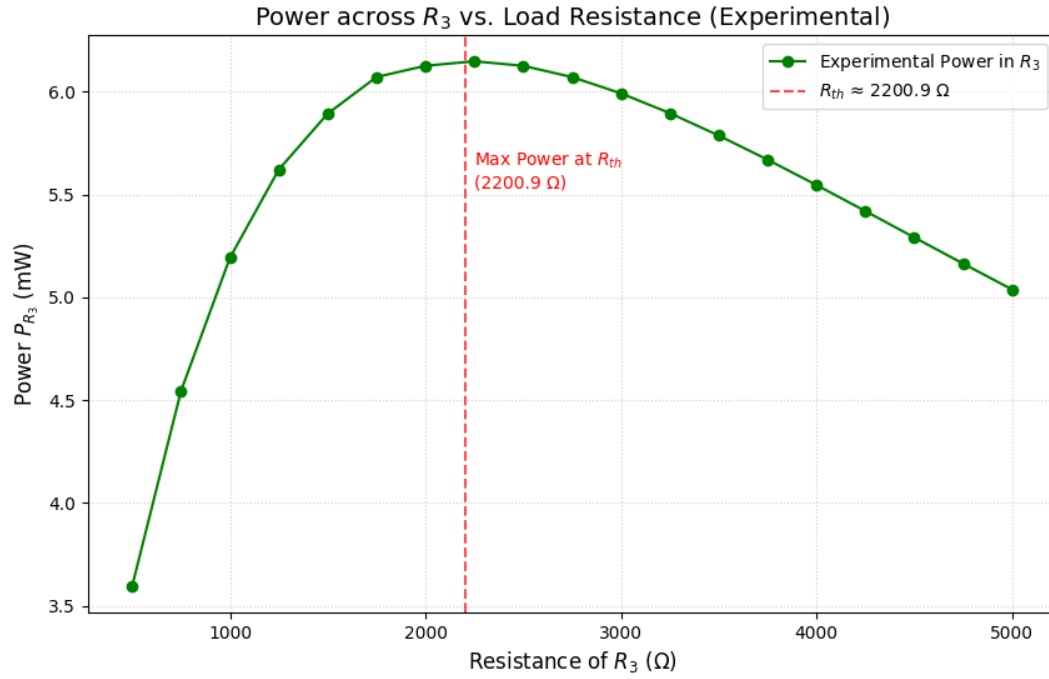


Figure 4: The experimental power delivered to the load resistor vs. load resistance with max power transfer occurring at $R_3 \approx 2200.9 \Omega$ (interpolated) which suggests that the actual resistor values for R_1 and R_2 differed from the theoretical values used.

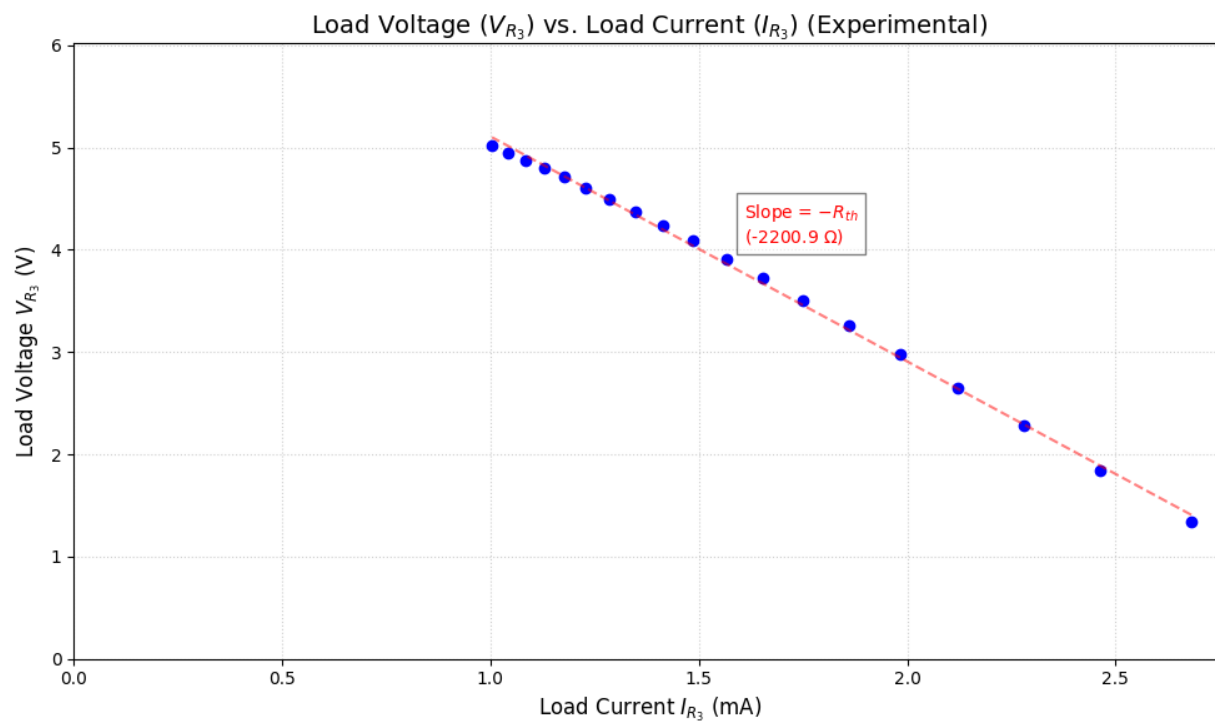


Figure 5: The experimental load voltage vs. load current data is plotted in blue with a linear fit shown in red with slope of magnitude approximately equal to the experimental Thevenin equivalent resistance R_{th} .

V. Discussion, Analysis and Concluding Statements

A. Slope of the load voltage vs. current plot

The slope of the load voltage vs. current plot in Figure 5 is approximately equal to the negative of the Thevenin equivalent resistance R_{th} of the source. This is consistent with the theoretical relationship derived from Kirchhoff's Voltage Law (KVL) which states that $V_{R_3} = V_{th} - I_{R_3}R_{th}$.

B. Load resistance that had the max power dissipation

The load resistance that had the maximum power dissipation was approximately 2200.9Ω , which is closest to the $R_3 = 2250\Omega$ value as seen in Figure 4. While close to the predicted value of 2779Ω , this shows that the Thevenin equivalent resistance of the source was likely lower than the theoretical value calculated in the Appendix.

C. Relationship to Thevenin equivalent resistance

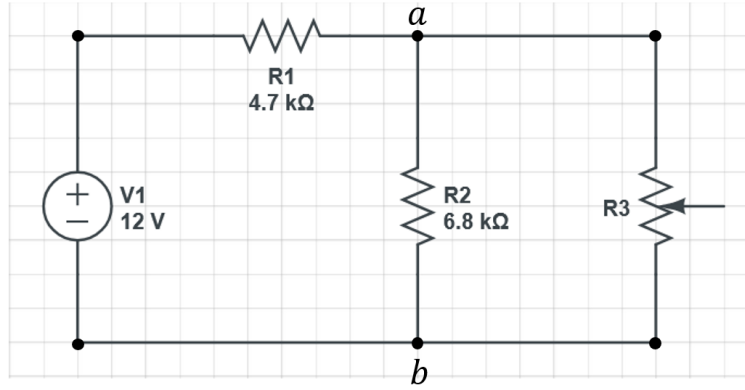
The maximum power transfer principle states that maximum power is delivered to the load when the load resistance equals the Thevenin equivalent resistance of the source. This is because we are able to redraw the circuit as a Thevenin equivalent circuit consisting of a Thevenin equivalent voltage source V_{th} in series with R_{th} and the load resistor R_3 . Since current is equal in a series circuit, voltage division says that the max voltage across the load resistor R_3 occurs when $R_3 = R_{th}$. Thus, the maximum power delivered to the load occurs when $R_3 = R_{th}$.

D. Conclusion

This experiment verified the maximum power transfer principle, showing that maximum power was delivered when the load resistance approximately equaled the Thevenin equivalent resistance of the source. Although the experimental Thevenin resistance differed from the theoretical value, the strong linear relationship in Figure 5 indicates that this discrepancy likely arises from differences between the actual resistor values and those assumed in calculations.

VI. Appendix

Calculations for Predicted Power Values



To find the Thevenin equivalent resistance R_{th} of the source, first remove the load resistor R_3 and turn off the voltage source v_1 (replace it with a short circuit). The remaining circuit consists of resistors R_1 and R_2 in parallel. Thus, the Thevenin equivalent resistance is given by:

$$R_{th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{[4.7 \text{ k}\Omega] \cdot [6.8 \text{ k}\Omega]}{[4.7 + 6.8 \text{ k}\Omega]} = 2.779 \text{ k}\Omega$$

Find the predicted voltage

Find the Thevenin equivalent voltage V_{th} by calculating the open-circuit voltage across nodes a and b. This can be done using voltage division:

$$V_{th} = v_1 \cdot \frac{R_2}{R_1 + R_2} = [12 \text{ V}] \cdot \frac{[6.8 \text{ k}\Omega]}{[4.7 + 6.8 \text{ k}\Omega]} = 7.095 \text{ V}$$

Using the Thevenin equivalent resistance, the circuit can be redrawn as a voltage source (V_{th}) in series with the Thevenin equivalent resistance (R_{th}) and the load resistor (R_3). Thus, the voltage across the load resistor R_3 can be calculated using voltage division:

$$V_{R_3} = V_{th} \cdot \frac{R_3}{R_{th} + R_3}$$

Find the predicted current

The current through the load resistor R_3 can be calculated using Ohm's law:

$$I_{R_3} = \frac{V_{th}}{R_{th} + R_3}$$

Find the predicted power

The power delivered to the load resistor R_3 can be calculated using the formula:

$$P_{R_3} = I_{R_3}^2 \cdot R_3$$